FINAL YEAR MINI PROJECT PRESENTATION

AI PREDICTIVE PERSONNEL STRESS AND WELFARE MONITORING SYSTEMS

A Smart Web Platform to Track, Predict, and Protect the Mental Health of Soldiers

STUDENT & PROJECT CREDENTIALS

NAME : DHARUN.M

YEAR : FINAL YEAR

DEPT : COMPUTER SCIENCE AND BUSINESS SYSTEMS

PROJECT TITLE : AI PREDICTIVE PERSONNEL STRESS AND WELFARE MONITORING SYSTEMS

Why is This Project Needed? (Real Military Challenges)

01

Hidden Stress in Field Duty

Soldiers deployed in remote border areas face high mental pressure and very little sleep, but their daily stress goes unnoticed until it becomes severe.

02

Manual Check-ups are Too Late

Existing psychological reviews happen only once in 3 or 6 months. This huge delay prevents officers from stopping critical mental breakdowns in time.

03

Soldiers Hesitate to Share Directly

Due to fear and hesitation, soldiers do not talk about their sadness openly, and commanding officers have no digital tool to spot high-risk jawans early.

What Did We Build to Solve This Problem?

01

Easy Daily Health Tracking

Build a simple website where every soldier can easily submit their daily sleep hours, duty stress, and heart rate in less than 30 seconds.

02

Automatic Color Alert (Triage System)

Automatically categorize soldiers into Red (High Risk), Yellow (Medium Risk), and Green (Safe) based on their daily health scores.

03

Instant Emergency Help & Calming Tool

Provide a 1-click Crisis SOS call button to Army Helplines and an interactive 4-7-8 breathing exercise to calm rapid heartbeats and panic.

How the System Works: 3 Roles & Cloud Setup

01

Three Clear User Roles

Strict division of duties: Soldiers log daily data; Welfare Officers monitor risk rosters and give counseling; Senior Admins manage overall Army stats.

02

Safe Cloud Database with Zero Data Loss

Developed using Python Flask and SQLite, connected to private Vercel cloud storage so that database records are permanently saved without losing data.

03

Works Smoothly on Mobile & Laptop

Designed with a clean mobile slide-out menu for smartphones and an executive two-column dashboard for laptops with zero horizontal scrolling.

SLIDE 05 | PERSONNEL MODULE

How a Soldier Uses the System Daily

01

4 Quick Daily Health Inputs

Soldier enters only four basic numbers: Sleep hours (0-12), Operational stress slider (1-10), Heart rate (bpm), and Squad morale (1-5).

02

Smartwatch One-Click Sync

A single 'Sync Smartwatch' button automatically fills sensor heart rate and sleep values so the soldier does not need to type manually.

03

Daily Streak & Achievement Badges

App rewards consecutive daily entries with positive streak counts and military badges (like 'Iron Mind') to encourage daily self-care habits.

SLIDE 06 | SMART HEALTH LOGIC

Automated Risk Triage & 7-Day Sleep Loss Forecaster

01

Automatic Red / Yellow / Green Tagging

If sleep is under 5 hours or stress is 8 or above, the app immediately tags the soldier in RED (High Risk) so officers can help right away.

02

7-Day Sleep Debt Calculation

App calculates total lost sleep compared to the 7-hour army baseline and creates a 0-100% Operational Fatigue Score.

03

Rest Alert to Prevent Duty Accidents

If a soldier's fatigue score crosses 65%, the system immediately recommends mandatory rest to prevent weapon accidents and duty mistakes.

1-Click Crisis SOS & 4-7-8 Breathing Calmer

01

1-Click Crisis SOS Button

A prominent red SOS button allows soldiers in severe distress to immediately call the Army Helpline (1902) or Tele-MANAS (14416).

02

Flashing Red Alert for Officers

When a soldier taps the SOS button, a flashing crimson red alert banner immediately pops up on the commanding officer's computer screen.

03

Tactical 4-7-8 Breathing Stabilizer

An animated circle guides the soldier: Inhale for 4s, Hold for 7s, and Exhale for 8s to quickly lower elevated heart rate and panic.

SLIDE 08 | COMMAND SUITE

Supervision Tools for Welfare Officers & Admins

01

Live Soldier Roster with Risk Filters

Welfare Officers can view all soldiers in their unit, filter by Red, Yellow, or Green, and immediately help soldiers in danger.

02

Confidential Counseling Diary

Officers can record private clinical counseling notes, save action steps, and schedule follow-up dates securely in the database.

03

Command Master PIN & Army Statistics

Senior Admins can view army-wide risk charts, check tamper-proof activity logs, and change the 4-digit signup security PIN anytime.

Tools Used & Safety Mechanisms

01

Reliable Modern Full-Stack

Built using Python Flask, SQLite database, Vercel Cloud Serverless platform, ReportLab for PDF exports, and Vanilla HTML/CSS/JS.

02

Strong Password & Recovery Security

Passwords are encrypted using PBKDF2 hashing. Soldiers can reset forgotten passwords using their Name and Service Unit without email/SMS.

03

Instant 3-Language Switcher

One-click language dropdown allows soldiers to switch between English, Tamil, and Hindi instantly for barrier-free comfort.

Project Summary & Next Steps

01

Successfully Tested & Live Online

The project is 100% completed and deployed live on Vercel at https://apps-nu-henna.vercel.app for real-world testing.

02

Fast, Friction-Free Experience

Removed all slow verification barriers so soldiers can log in directly and submit their health records in under 30 seconds.

03

Future Plans: Real Smartwatches & Voice AI

Future upgrades include direct Bluetooth pairing with actual military smartwatches and AI voice sentiment analysis during counseling.